CUSTOMER RETENTION & CHURN ANALYSIS

Final Data Analytics Project Report

Report Date: 06-11-2026

Project Folder: C:\Users\gowth\OneDrive\Documents\FUTURE_DS_02

1. Project Overview

This project analyzes customer churn patterns in a telecommunications company using the Telco Customer Churn dataset. The objective is to understand customer retention, identify groups with higher churn rates, and suggest strategies to improve customer loyalty.

2. Dataset Details

Metric	Value
Total Customers	7,043
Total Columns	21
Churned Customers	1,869
Retained Customers	5,174

3. Tools and Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly and Jupyter Notebook.

4. Key Performance Indicators

KPI	Result
Total Customers	7,043
Churned Customers	1,869
Retained Customers	5,174
Overall Churn Rate	26.54%
Average Monthly Charges	$64.76
Average Tenure	32.37 months

5. Key Findings

• Month-to-month contract customers have a 42.71% churn rate.

• Electronic check users have the highest payment-method churn rate at 45.29%.

• Fiber optic customers have a churn rate of 41.89%.

• Senior citizens have a churn rate of 41.68%.

• Customers who churned had a lower average tenure than retained customers.

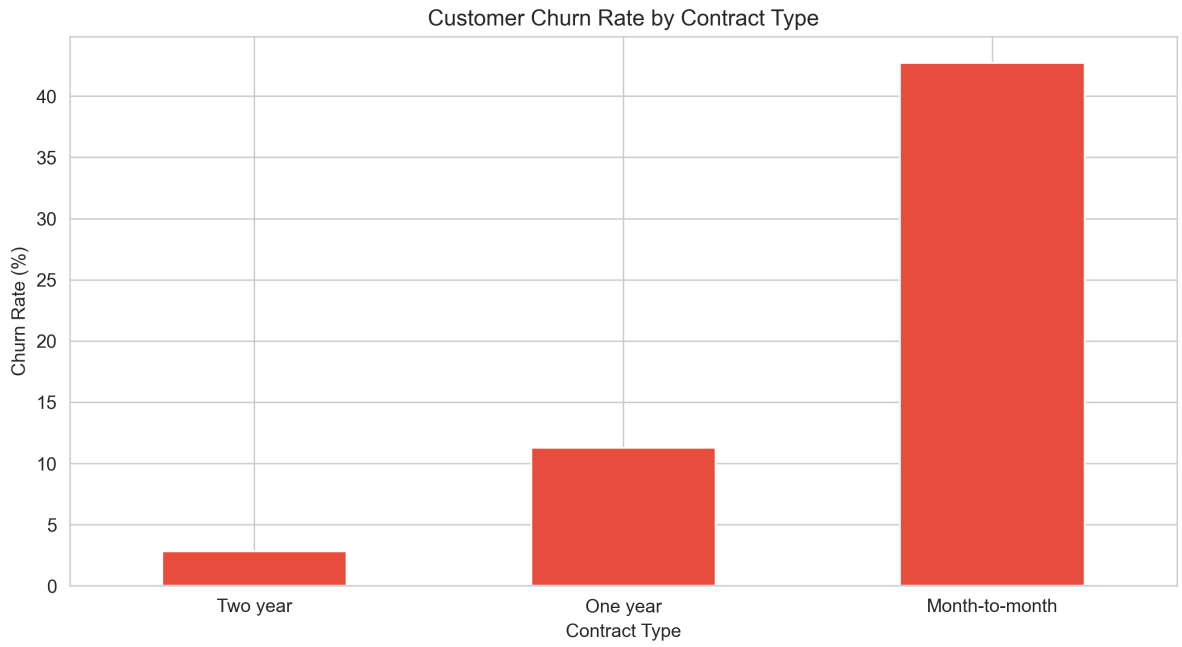
• Customers who churned had higher average monthly charges.

6. Recommendations

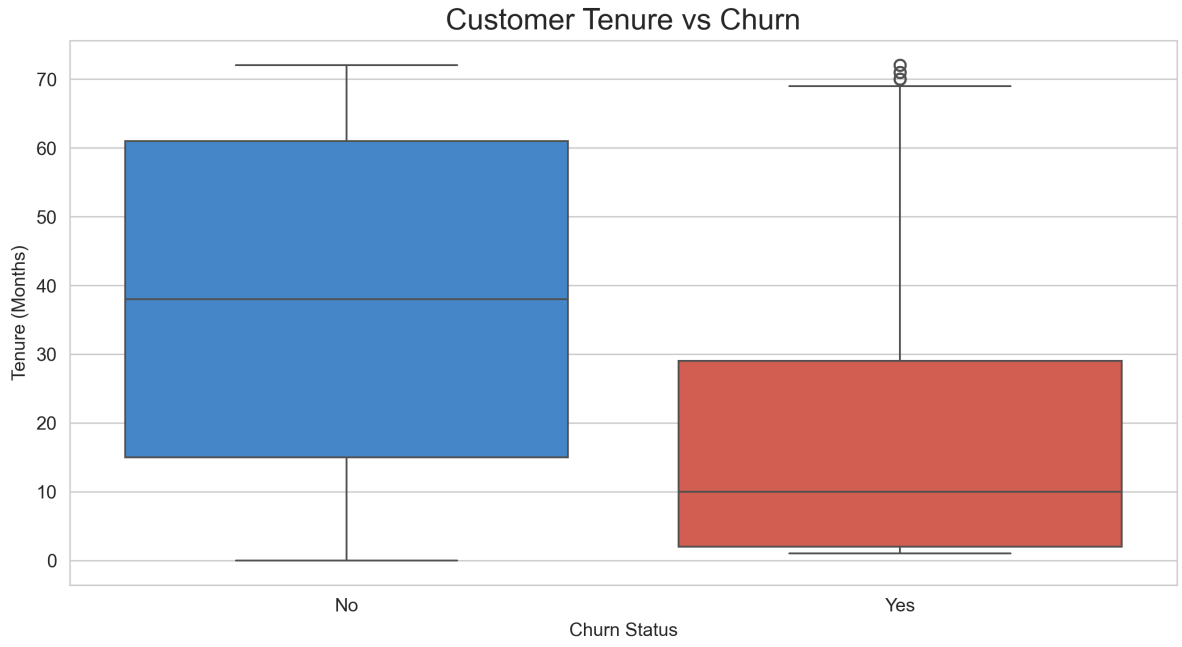
- Provide retention offers to month-to-month customers.
- Review pricing and service quality for fiber optic customers.
- Encourage convenient automatic payment methods.
- Improve onboarding and support during the first year.
- Offer personalized assistance to customers at higher churn risk.

7. Data Visualizations

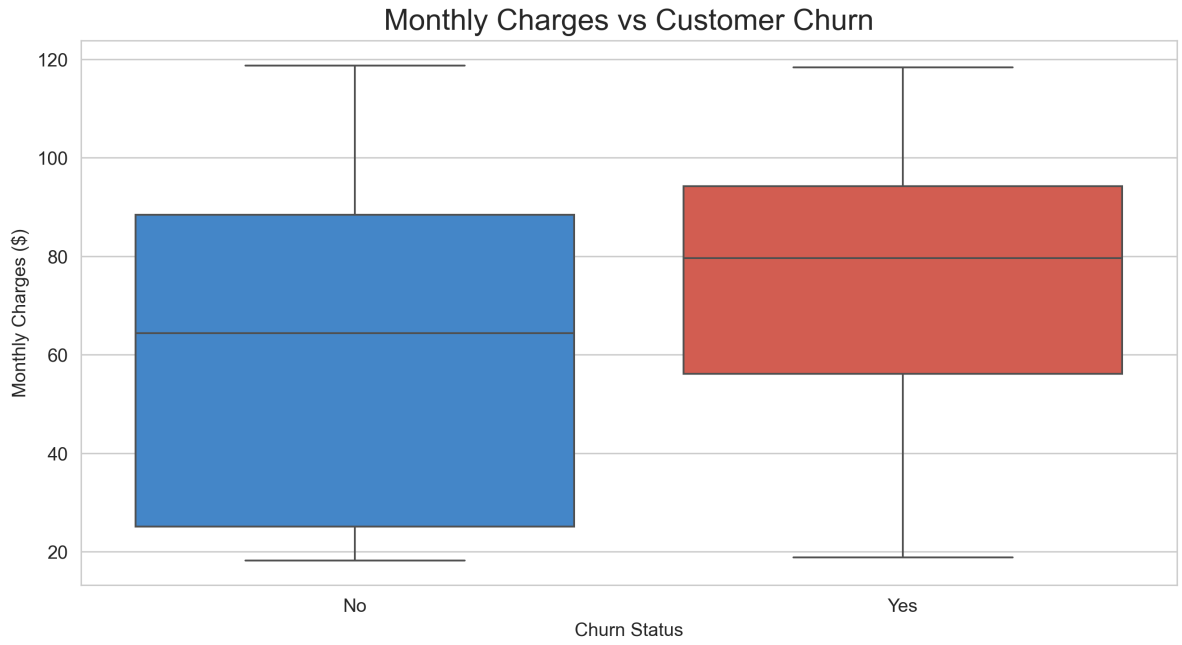
Churn By Contract



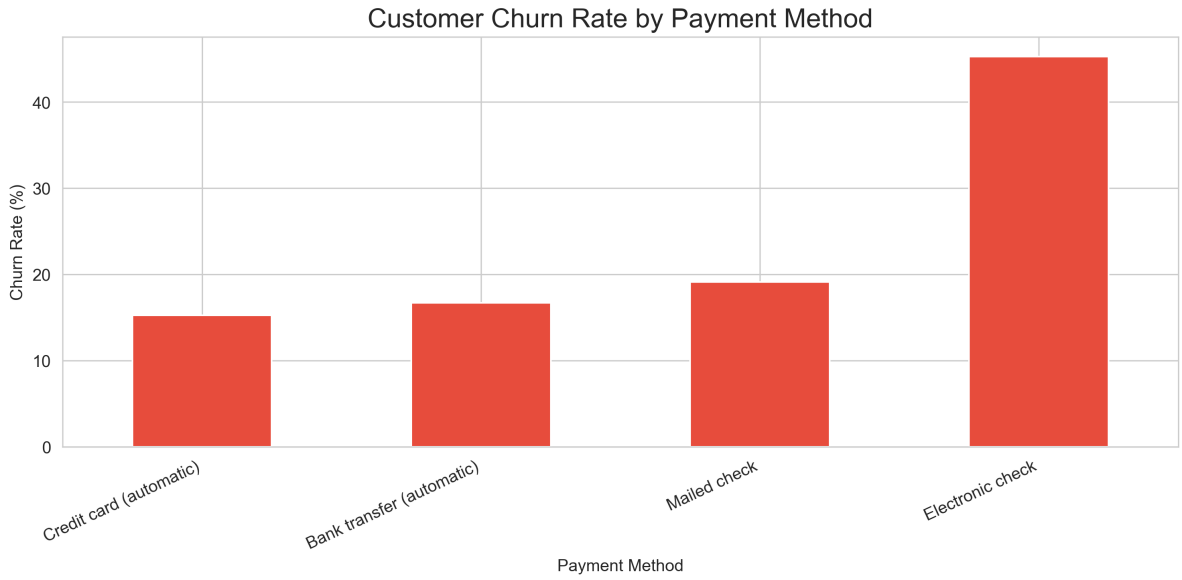
Tenure Vs Churn



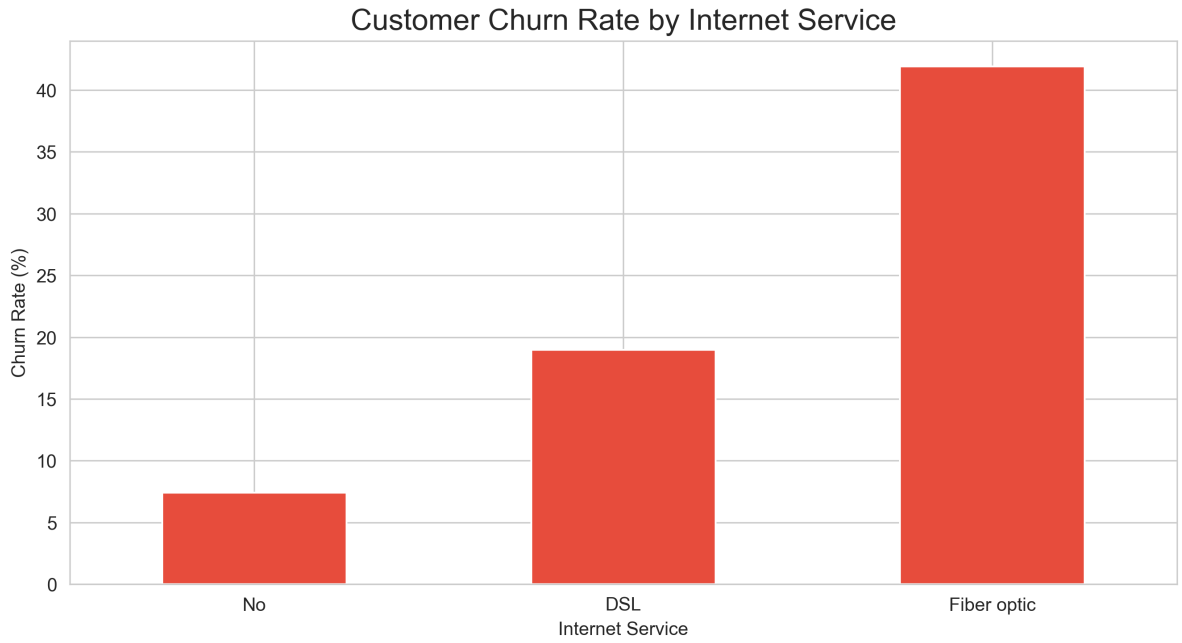
Monthly Charges Vs Churn



Churn By Payment Method



Churn By Internet Service



Churn Risk Factors



8. Interactive Dashboard

Interactive dashboard file: dashboard/Customer_Churn_Interactive_Dashboard.html

9. Conclusion

The analysis identifies important customer groups associated with higher churn rates. Month-to-month contracts, electronic check payments, fiber optic service and shorter customer tenure are notable patterns. Targeted retention strategies and improved customer support may help reduce churn. These findings represent associations in the dataset and do not establish causation.